

10
10

- LLN: Let a distribution have some mean μ and finite variance $\sigma^2 < \infty$, then

$$\lim_{n \rightarrow \infty} P\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu\right) = 0$$

Section 10.6

1. $H_0: \mu_1 = \mu_2$ ✓

$H_a: \mu_1 \neq \mu_2$

$$t = \frac{\bar{y}_1 - \bar{y}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{74 - 71}{\sqrt{\frac{61}{50} + \frac{100}{50}}} = 1.517$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}} = \frac{(1.62 + 2)^2}{\frac{1.62^2}{49} + \frac{2^2}{49}} \approx 96.93 \approx 97$$

$$p\text{-value} = 2 \cdot P(T > |t|) = 2 \cdot P(T > 1.517) = 2 \cdot \text{tcdf}(1.517, 1E99, 97) \approx .118$$

b. Since $.118 > .05$, we fail to reject the null hypothesis and conclude there's no evidence to suggest a difference between true mean score for 2 independent sets of elementary school children when taught to read

53. $H_0: \mu_1 = 3.8$ ✓

$H_a: \mu_1 < 3.8$

$$t = \frac{\bar{y} - \mu_0}{s/\sqrt{n}} = \frac{3.6 - 3.8}{1.1/\sqrt{30}} = -.996$$

$$p\text{-value} = P(T < -.996) = \text{tcdf}(-1E99, -.996, 29) \approx .164$$

b. Since $.164 > .05$, we fail to reject the null hypothesis and

conclude that there's not enough evidence to conclude that a decrease in the mean FVC over the decade is the true

c $H_0: \mu_1 = 3.1$

$H_a: \mu_2 < 3.1$

$t = \frac{\bar{Y} - \mu_0}{s/\sqrt{n}} = \frac{2.7 - 3.1}{1.2/\sqrt{30}} \approx -1.826$

p-value = $P(T \leq -1.826)$

$= \text{tcdf}(-1.826, -1.826, 29) \approx 0.039$

Since $0.039 \leq 0.05$, we reject the null hypothesis and conclude there's enough evidence that the fitness program

reduces FVC drop for women below 3.1 dl/decade. true mean drop in FVC over the decade for women on the physical fitness program is less than 3.1 dl.

$H_0: p = 0.85$

$H_a: p > 0.85$

$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.96 - 0.85}{\sqrt{\frac{0.85 \cdot 0.15}{300}}} \approx 5.34$

p-value = $P(Z > 5.34)$

$= \text{normalcdf}(5.34, 1E99, 0, 1) \approx 0$

a. Since $0 < 0.01$ we reject the null hypothesis and conclude that there's enough evidence that proportion of right-handed CEOs is higher than general population.

55. $H_0: p = 0.05$

$H_a: p < 0.05$

$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{\frac{45}{1124} - 0.05}{\sqrt{\frac{0.05 \cdot 0.95}{1124}}} \approx -1.533$

p-value = $P(Z \leq -1.533)$

$= \text{normalcdf}(-1E99, -1.533, 0, 1)$

≈ 0.0628

At 1% level, $0.0628 > 0.01$ so we fail to reject H_0 and conclude there's not enough evidence verification system helps.